

State of Developer Ecosystem 2024

Language Trends, Satisfaction, and Strategic Shifts

Insights from JetBrains' survey of 23,262 developers worldwide

23,262

Developers Surveyed

20

Languages Tracked

8

Years of Data

JavaScript peaked at 70% (2020) but has declined to 61%. Python rose from 32% to 57%, nearly closing the gap.

Python Closes the Gap: 8 Years of Language Usage Shifts

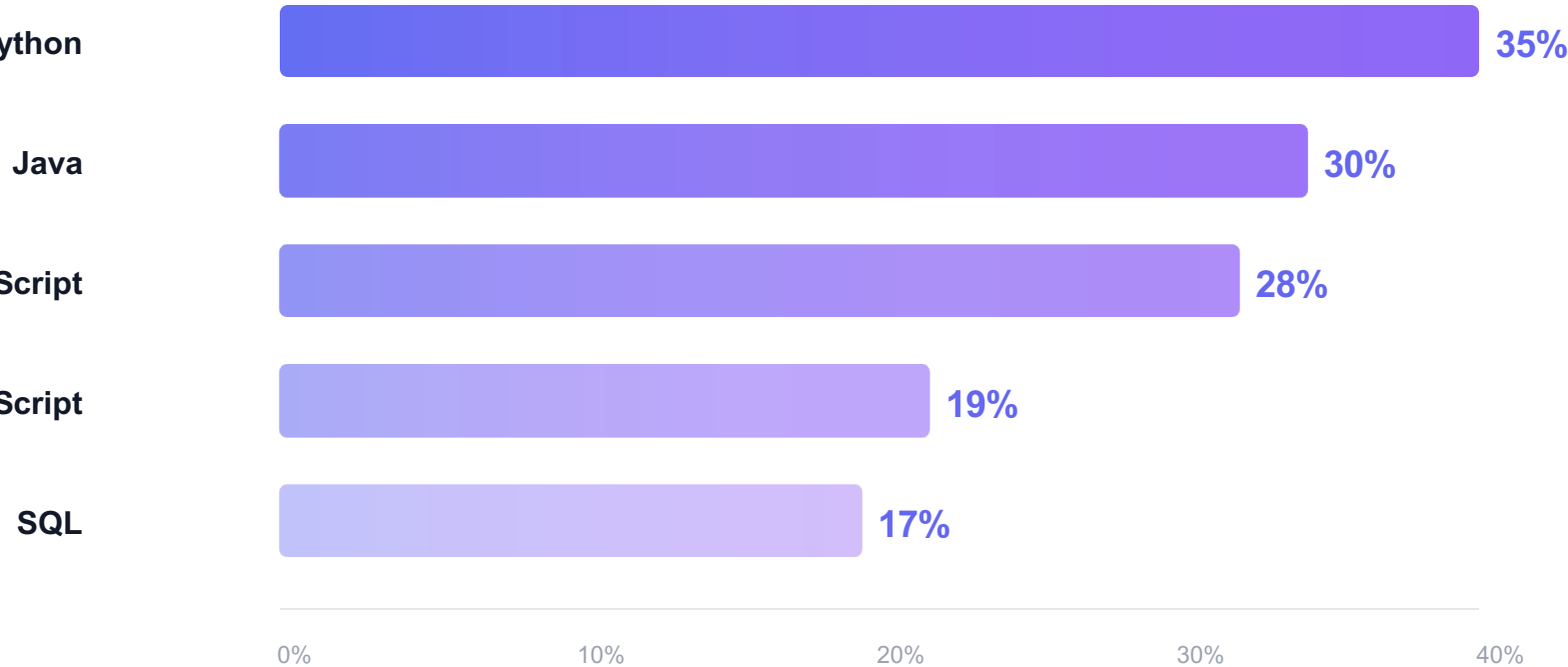
Usage trends 2017-2024 · Percentage of developers using each language

Language	2017	2020	2022	2024	Change	Trend
JavaScript	65%	70%	65%	61%	-4%	▼ Declining
Python	32%	55%	53%	57%	+25%	▲ Strong growth
HTML/CSS	60%	61%	54%	51%	-9%	▼ Gradual decline
SQL	42%	56%	49%	48%	+6%	— Plateaued
Java	47%	54%	48%	46%	-1%	▼ Slow decline
TypeScript	12%	28%	34%	37%	+25%	▲ Tripled
PHP	30%	27%	20%	17%	-13%	▼ Significant decline
Go	8%	19%	19%	18%	+10%	— Plateaued after growth
Rust	—	7%	9%	11%	+4%	▲ Steady growth

Despite JavaScript's overall lead, Python is now the #1 primary language — reflecting AI/ML dominance.

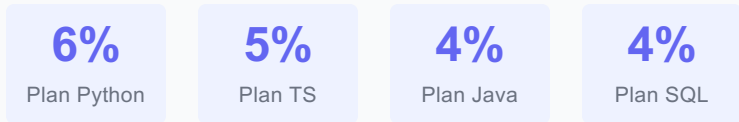
Python Is Now the #1 Primary Language at 35%

When asked what language developers primarily use · Compare with adoption intent below 6% for all languages



Ecosystem Lock-in Effect

Adoption intent remains below 6% for all languages, revealing strong ecosystem lock-in.

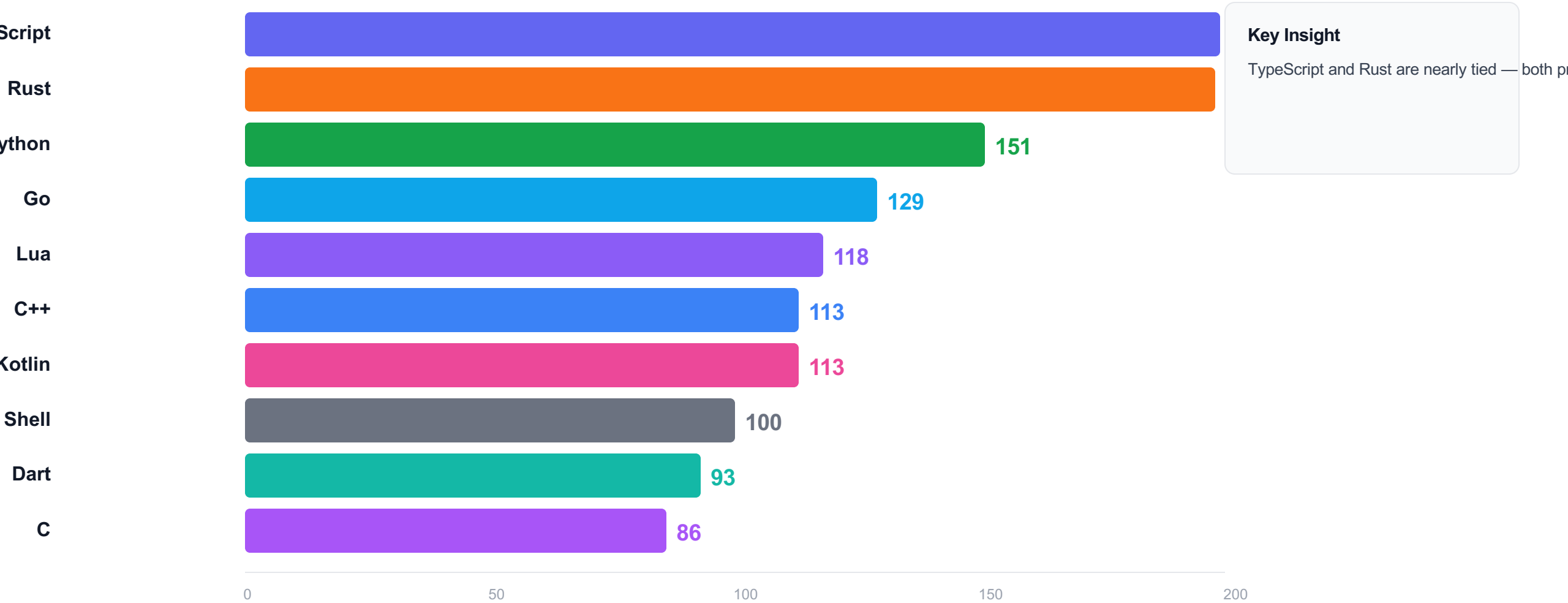


Developers rarely switch their primary stack. The combination of expertise investment and ecosystem lock-in creates a high barrier to entry for new languages.

Type-safe, modern-tooling languages dominate — TypeScript and Rust score nearly identically as most loved.

TypeScript and Rust Top Developer Satisfaction, Nearly Tied at 199 vs 198

Developer satisfaction score (higher = more loved) · Top 10 languages by satisfaction



Language choice is driven by domain expertise, not general preference — each language owns a specific domain.

Each Language Owns a Domain: AI Belongs to Python, UI to TypeScript

Top language choice by application domain · Percentage of developers using the leading language

Application Domain	Leading Language	Share	Why It Leads
AI / Machine Learning	Python	31%	Dominant in data science, ML frameworks, AI/ML ecosystem
Data Science & Analytics	Python	28%	Pandas, NumPy, Jupyter ecosystem; data processing
Data Processing & Analytics	Python	17%	ETL pipelines, data wrangling, analytics tools
User Interface	TypeScript	62%	React, Vue, Angular — modern web frameworks
APIs & Backend Services	Java	49%	Enterprise Spring ecosystem, Android, microservices
Application Logic & Workflows	Java	55%	Strong typing, enterprise patterns, concurrency
System Tools & Components	C++	27%	Performance-critical, hardware interfacing, rendering

What This Means for Your Technology Strategy in 2025

STRATEGIC PRIORITY #1

Invest in Python for AI/ML Teams

Python is the undisputed leader in AI/ML and data science (31% AI, 28% data science). But TypeScript dominates UI development (62%) and scores highest in developer satisfaction. Migrate Java

STRATEGIC PRIORITY #2

TypeScript Is the Future of Frontend

TypeScript dominates UI development (62%) and scores highest in developer satisfaction. Migrate Java

STRATEGIC PRIORITY #3

Rust and Go Are Growth Plays for Infrastructure

Rust (198 satisfaction) and Go (129, plateaued at 18%) are the languages for modern infrastructure. PHP dropped from 30% to 17% (2017-2024); Ruby fell from 10% to 4%. Long-term decline signals shrink

HIRING WARNING #4

PHP and Ruby Hiring Pools Are Shrinking

PHP dropped from 30% to 17% (2017-2024); Ruby fell from 10% to 4%. Long-term decline signals shrink

RETENTION STRATEGY #5

Developer Satisfaction Correlates with Type Safety and Modern Tooling

Bottom Line: 2025 technology strategy should prioritize Python for AI/data, TypeScript for frontend, and Rust/Go for infrastructure — while planning migrations away from PHP/Ruby ecosystems.

Source: JetBrains State of Developer Ecosystem 2024 (n=23,262) · CONFIDENTIAL